Hamza Aamer

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EDUCATION

Karlsruhe Institute of Technology (KIT) **Oct. 2025 – Aug. 2027 (expected)**
M.Sc. Computer Science, Human-centred Machine Intelligence *Karlsruhe, Germany*

National University of Computer & Emerging Sciences (FAST-NUCES) **Aug. 2021 – June 2025**
B.Sc. Computer Science, Gold Medalist, Rector's List of Honor *Islamabad, Pakistan*

WORK EXPERIENCE

Adept Tech Solutions **Nov. 2024 – Aug. 2026**
AI Engineer & Product Lead

- Spearheaded development of advanced data synthesis and analysis systems using PySpark, statistical modeling, and computer vision, achieving **10x** improvement in data generation accuracy and automation of complex workflows. Key achievements include:
  - Engineered proprietary **DNA Signature Technology** spanning flat file and relational database synthesis: **flat file DNA** extracts statistical fingerprints via multiple binning strategies with distributed correlation signatures supporting time series analysis and PII anonymization; **Database DNA** captures three distinct DNA types (Table, Relationship, Junction) using a hybrid multi-type correlation engine with adaptive binning, degree-first child generation, joint parent conditioning for N:N relationships, multi-level fallback chains, and encrypted signature packaging, reducing terabyte-scale datasets to compact portable signatures while maintaining **99.5%** statistical fidelity and full cross-table relationship preservation.
  - Designed **Rule-Based Synthesis Framework** for both flat files and relational databases, layering configurable business logic on top of statistical generation through multi-layered scopes with compound conditions and multiple effect methods, implementing two-phase generation with topological dependency ordering, multi-hop join-path condition evaluation, filtered FK resolution with uniqueness tracking, and batch cross-table modifications, ensuring enterprise constraint satisfaction.
  - Implemented **13+ specialized data engineering modules** including intelligent class balancing that automatically calculates optimal scaling factors for imbalanced datasets, outlier detection, time series standardization, and precision correction, processing datasets exceeding **10M** rows while reducing synthesis time from hours to minutes.
- Led deployment of production-ready CLI tools with distributed computing optimization across Spark clusters, enabling seamless pipeline execution of Ascend workloads on multi-node deployments.

Antematter **Jan. 2023 – Nov. 2024**
AI Engineer

- Contributed to the development of various Multi-Agent Frameworks using orchestration frameworks, including CrewAI, LangChain, ELIZA, and Phidata, significantly boosting company outreach by **10x**. Key achievements include:
  - Developed **Ant-AI**, a genetic multi-agent framework built with CrewAI to optimize system prompts for low-parameter LLMs, reducing input prompt ambiguity by **70%** and fully automating prompt engineering.
  - Designed supervisory agents within crew environments capable of autonomously generating additional AI agents with function-calling capabilities, enabling tailored collaboration and integration of individual strengths toward collective tasks.
  - Engineered the **First Genetic Agent Framework** capable of generating and iteratively refining results through collaborative Synthesis and QA agents. Developed specialized Agentic Mutation and Crossover Crews that enhance adaptability and accuracy.
  - Designed and deployed virtual clones of real personalities using ELIZA, enabling autonomous control over social media platforms to expand online presence and automate lead generation for ICP interactions, increasing online engagement by **8x**.
- Took part in a project with **Nedge Computing**, in collaboration with **NVIDIA**, consisting of creating virtual cloud systems with divided vGPU access of Nvidia A16 GPUs.

Afiniti Software Solutions Private Limited **June. 2020 - Aug. 2021**
Project Trainee

- Engaged in software development activities, including the design and implementation of a Natural Language Processing **(NLP)** and **Speech to Text** system powered by Artificial Intelligence.
- Actively participated in client interactions, ensuring their requirements were met with AI-enhanced solutions.

RESEARCH EXPERIENCE

Chair for Embedded Systems (CES/ITEC), KIT **Apr. 2026 – present**

Research Assistant (HiWi), Prof. Dr. Jörg Henkel *Karlsruhe, Germany*

- Research on **ML-driven GPU resource management for LLM inference**, extending the **gpu_ext** eBPF/struct_ops framework to expose scheduling, memory and frequency levers on real hardware.
- Designed and validated a **channel-disable scheduling policy** separating latency-critical from best-effort workloads (up to **15x** latency improvement), and evaluated unified memory prefetch and oversubscription policies (**5-12%** throughput gain).
- Reproduced published **kernel-level DVFS** results and characterised GPU frequency-switching cost via **NVML**, separating API latency from physical clock transition time to establish the feasible granularity of frequency control.

TECO, Pervasive Computing Systems, KIT **Apr. 2026 – present**

Research Assistant (HiWi), Prof. Dr. Michael Beigl *Karlsruhe, Germany*

- Full-stack development on **edge-ml**, a microservice platform (FastAPI, React, Docker) for training and deploying ML models on wearable and IoT sensor time-series.
- Implemented on-device model export via **Google ExecuTorch**, compiling trained PyTorch classifiers with preprocessing into `.pte` artifacts for mobile and embedded targets, with auto-generated Kotlin integration code.

DataInsight Research Lab, FAST-NUCES **May 2023 – Dec. 2023**

Research Assistant *Islamabad, Pakistan*

- Contributed to a **$90,000** machine learning research project on document understanding and text recognition.
- Built a **YOLO**-based detection and image processing pipeline for handwritten and printed text recognition, improving dataset acquisition speed **2.5x**, and trained a OCR model for text extraction from medical prescriptions at **84%** accuracy.
- Replaced a multi-component OCR and language model stack with an end-to-end transformer approach (**DocParser**, **Donut**), reducing compute requirements by **50%** while producing structured output directly.

Notable Projects:

Ascend: Enterprise Data Engineering Framework (04/2025) **Python, PySpark, PostgreSQL, MySQL**

- Modular data engineering framework with proprietary DNA signature technology for synthetic data generation preserving complex relationships. Includes 30+ specialized modules for data analysis, curation, feature engineering and data transformation.

Artisan Roofing: AI-Powered Aerial Analysis Platform (03/2025) **Python, Flask, YOLOv8, Open3D, QGIS**

- Professional roof measurement system integrating satellite imagery with LiDAR point cloud processing to automatically classify roof components and generate contractor-grade reports. Implements multi-stage geometric algorithms delivering comprehensive analysis in under a minute.

Ant-AI: Multi-Agent Prompt Optimization Framework (11/2024) **Python,CrewAI, Streamlit, Poetry**

- Framework for optimizing task prompts using multi-agent systems and genetic algorithms. Implements dynamic agent roles for iterative refinement, employing reasoning tools like Tree of Thoughts and structured prompt tagging for enhanced LLM input generation.

GenSynth: Adversarial Image-to-Image Translation (07/2024) **Python, PyTorch**

- Generative adversarial framework utilizing CycleGAN and Conditional GANs for bidirectional face sketch-to-image translation, ensuring high-precision synthesis through cycle-consistency and pixel-wise L1 loss.

NeuroCart: A Cognitive Shopping Fusion Framework for the Future (10/2023) **Python,Java,Kotlin,XML,Collab**

- Mobile application merging online and physical shopping experiences, enabling users to employ a trained vision model for real-time item detection using their device's camera.

Multi-Core Neural Network Operating System in C (05/2023) **C**

- A multi-core operating system to implement a neural network architecture using separate processes and threads, enabling efficient utilization of multiple cores for parallel processing.